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ABARES and its predecessors have been providing commodity market forecasts for Australia's agricultural sector since the 1940s, as part of a wider suite of evidence-based analysis and advice to policymakers and industry stakeholders. The operational context of ABARES and the industries we serve has changed a great deal over this period, and there are no signs that future changes will be slower or less profound.

This paper explores the future of public sector agricultural forecasting in the context of the past evolution of ABARES work. It identifies a range of challenges and opportunities for people involved in public sector forecasting, including the usual suspects of globalisation and global change, rapid improvements in information and communication technology, the emergence of big data (in all its forms), and new technology-enabled possibilities for interaction and co-production of knowledge.

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Summary

Australian governments since 1945 have provided commodity market forecasts for Australia's agricultural sector. Since then the operating and policy context of Australian agriculture has changed dramatically. Public investment in agricultural forecasting fell from the 1990s onwards as technology made forecasting more efficient and as agriculture's falling share of the economy reduced its priority within government. The acceleration of global change into the 21st century brings into question the future role of public sector forecasting in Australian agriculture and what these forecasting services should look like into the future.

Public sector forecasts for Australian agriculture have almost exclusively been provided by the Bureau of Agricultural Economics (BAE, 1945–1987) and its successors, the Australian Bureau of Agricultural and Resource Economics (ABARE, 1987–2010) and the Australian Bureau of Agricultural and Resource Economics and Sciences (ABARES, from 2010). The idea of creating a BAE surfaced during World War II and was initially driven by the need to provide an evidence base for policy development. Prior to the war, public investment in agricultural economics and market analysis was limited to research that supported sporadic industry inquiries.

Government commitment to forecasting began during World War II

The Australian Government's commitment to providing forecasting services for agriculture emerged from 1943 onwards as the scale of dismantling wartime policy became clear. This was intensified by anticipation of the problems associated with resettling returned soldiers with limited farming experience onto farms acquired by governments.



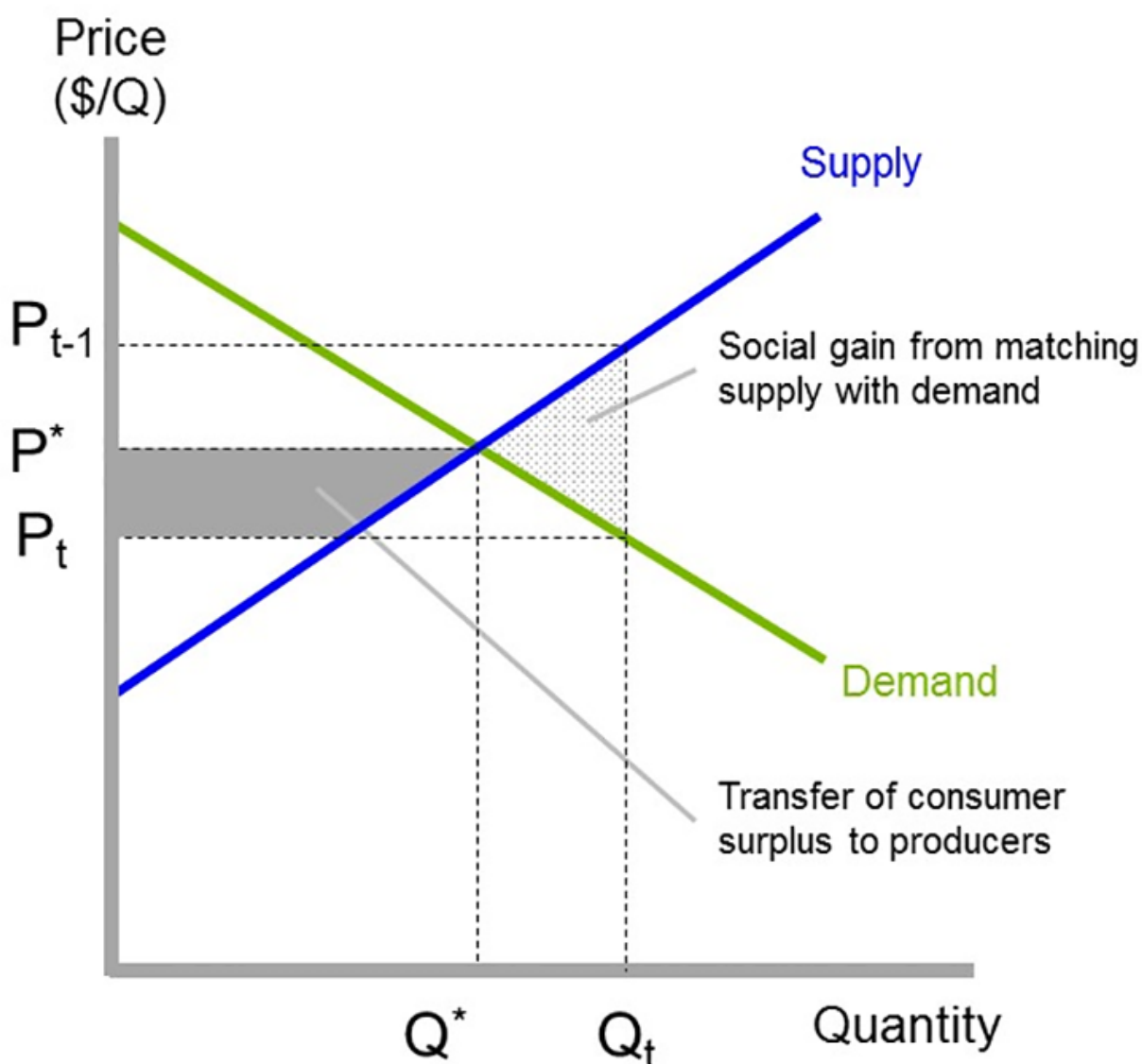
Digital Image ID 2606, An early settler, Beerburrum, December 1916

Many soldier settlement farms established after World War I had failed, partly as a result of naive commodity forecasts. The bureau's flagship publication, the *Quarterly Review of Agricultural Economics*, was first published in January 1948 and included quarterly updates of expected trends in agricultural markets.

Australia's BAE owed much in its original design to similar institutions that had already been established in the United States and Canada. Governments in the United States and Australia both saw the public provision of production and price forecasts as a means of addressing a perceived imbalance in information between farmers and traders. In Australia, the public provision of forecasting services was complemented by statutory marketing. Statutory marketing was designed to provide farmers with countervailing market power in their negotiations with traders. However, it also removed responsibility for marketing from farmers and incentives to innovate in response to market signals.

Statutory marketing was just one manifestation of a high level of post-war government involvement in agriculture that sustained policy demand for agricultural forecasts by central government agencies. BAE's forecasting services expanded from the 1960s to the 1980s in terms of commodity coverage and methodological capability, with public investment in forecasting services peaking in the mid-1980s. In 1981–82, 65 staff were involved in commodity and marketing economics research and a further 42 in collecting and processing commodity data. The commitment of resources during the 1980s allowed experimentation with increasingly sophisticated structural and programming models.

Forecasts provide economic value through market efficiency



Economists have always emphasised the economic value that forecasts provide through the efficient operation of markets, with benefits to consumers and producers. They also tend to be equivocal about whether and to what extent these services should have been exclusively funded by governments. Reliable forecasts can help reduce the uncertainty that farmers face when committing resources to production well in advance of knowing demand at the time of marketing. Consumers benefit through the timely availability of high quality and reasonably priced food. This provides an incentive for both consumers and producers to contribute to the cost of forecasts. Economic thinking can also be used to understand the value derived from using forecasts to make markets more equitable, and to support policy development.

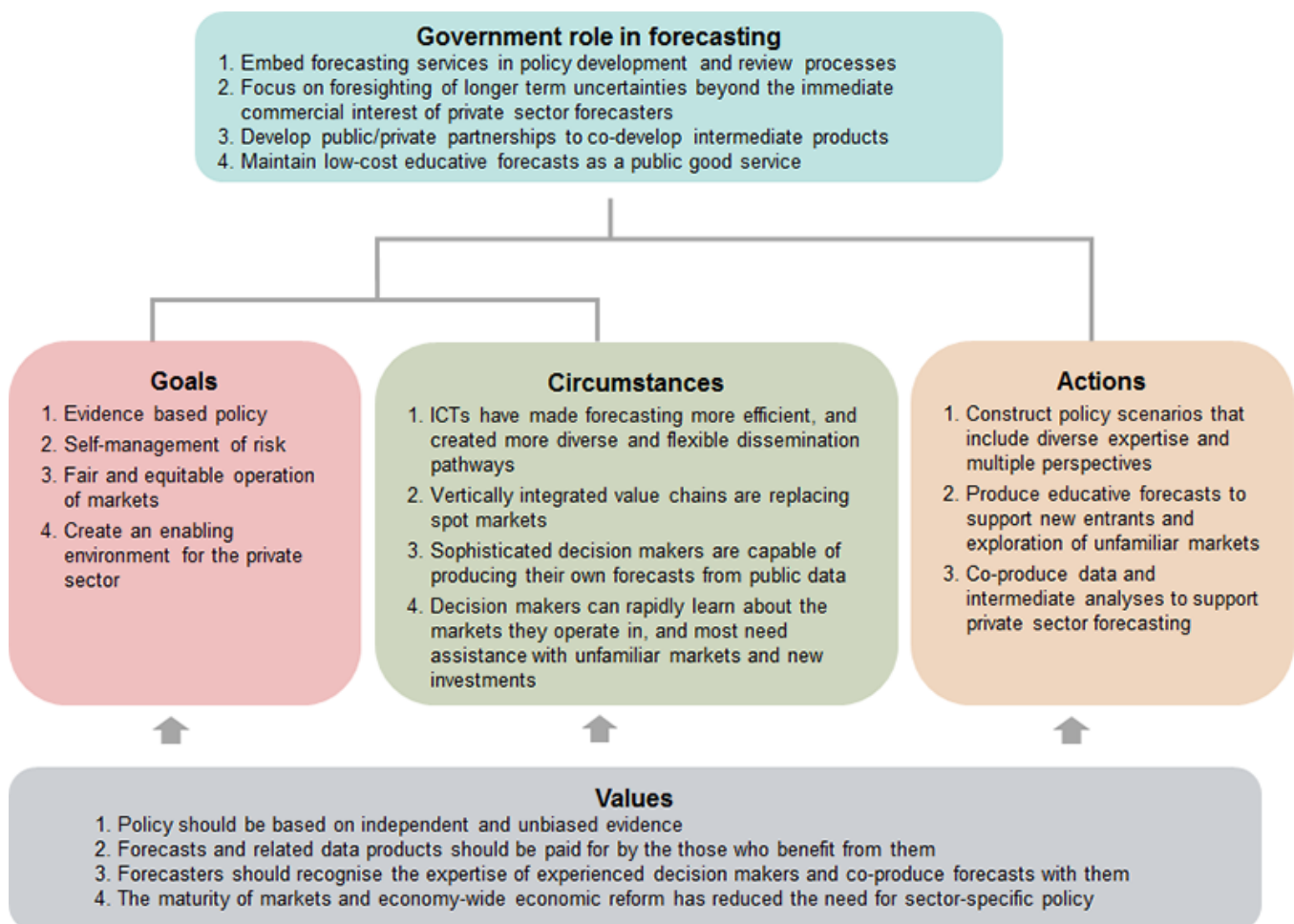
Policy demand for agricultural forecasts remains strong into the 21st century. Although the emphasis on industries and issues changes constantly, the basic policy applications of ABARES forecasts have changed little since 1945. Forecasts and related market information are used to respond to stakeholder concerns and form policy responses to emerging issues, and continue to be used by central agencies for

macroeconomic forecasting. ABARES forecasts remain essential for policy applications where independence from industry and markets is essential.

Globalisation has changed policy arguments for public forecasting

Beyond this policy role, globalisation has dramatically changed the policy arguments shaping the future public provision of forecasting services for Australian agriculture.

The growing sophistication of forecast users and development of interactive web-based technologies means that public sector forecasting services should focus on providing intermediate data and analyses that users can recombine to produce their own forecasts. In government, this would build the capacity of policy advisers to tailor advice in response to emerging issues. In the private sector, this would boost the value of public good data and analyses that consultants and others use to tailor commercial forecasts.



The growing expertise of users increases their independence from expert forecasters and creates a store of knowledge that forecasters should draw on. This will require ABARES to evolve from an expert-centric institution to one via which expert forecasters engage meaningfully with diverse groups of forecast users. This is likely to involve greater use of social media and interactive web-technologies that support innovative approaches to consensus-based foresighting of deeply uncertain future market scenarios.

Future forecasting services need to evolve to meet changing user needs

Globalisation has fundamentally altered relationships between farmers and other participants along vertically integrated value chains. Former adversaries in spot markets are now mutually dependent on one another in vertically integrated value chains, which potentially improves the flow of information between them. This means that future forecasting services can no longer focus exclusively on farmgate prices, but need to inform the creation of value along vertically integrated value chains.

Public sector forecasting services also need to evolve a complementary focus on foresighting, using scenarios built through consultation that draws on diverse types of expertise and multiple perspectives. Scenario analysis that anticipates the future vulnerability of agricultural industries would support proactive policy design, particularly for significant and uncertain step-wise change that is difficult for markets and industries to assess and adapt to. It would also support the investigation of significant new investments and forays into unfamiliar markets by increasingly sophisticated forecast users.

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